financiële wiskunde : oefeningen

lecture 2 : 24/02/2026

Show that $E[X^4] = 3\sigma^4$, $X \sim N(0, \sigma^2)$

$$X \sim N(0, \sigma^2) \Rightarrow Z = X/\sigma \sim N(0,1)$$

$$\Rightarrow \sigma Z = X$$

$$\Rightarrow E[X^4] = E[(\sigma Z)^4]$$
$$= \sigma^4 E[Z^4]$$
$$= 3\sigma^4$$

$$\left(\begin{array}{l} Z \sim N(0,1) \\ \quad \hookrightarrow E[Z^4] = \left.\dfrac{d^4 M_Z(t)}{dt^4}\right|_{t=0} \\ M_Z(t) = e^{t^2/2} \\ \quad = 1 + \frac{1}{2} \end{array} \right)$$